

Setting Up Your Development Environment

Learning to program requires that you actually DO lots of programming – not just read about it, hear about it, or watch someone else do it, but actually do it yourself. That means that you'll need to set up a programming environment so you can learn all the stuff in this book by actually slinging code. For the topics covered in this book, you'll need to install Visual Studio Community 2019 and the Unreal game engine. Doing this will let you use Visual Studio to write C++ console applications and will also let you write C++ scripts in Unreal as you develop Unreal games. We'll discuss how to install each below.

A.1. Visual Studio Community 2019 (Windows only)

Microsoft's commercial Integrated Development Environments (IDEs) are called Visual Studio, and they come in various capability (and cost) levels. Because there are lots of people who want to program as a hobby rather than as a commercial endeavor, Microsoft also distributes free IDEs. We used the free Visual Studio Community 2019 IDE to develop all the code for this specialization. Although the source code provided on Coursera can be used in any IDE, if you have Visual Studio Community 2019 you can just open up and use the projects as is.

NOTE: Mac users can't do C++ development using Visual Studio. Instead, Mac users will need to install Xcode as their IDE. Instructions for doing that are provided at the end of this document.

Here's how you can get this IDE installed:

1. Download Visual Studio Community 2019 at <http://www.visualstudio.com/>.
2. Run the install file you downloaded.

Be sure to check the following items in the Workloads area of the installer:

Desktop & Mobile: Desktop development with C++
Gaming: Game development with C++

You should also check the following in the Individual components area of the installer:

Code tools: Help Viewer

You can certainly select other Workloads and Individual components if you'd like, but we just installed the three items above. After selecting those items, click the Install button and wait patiently while everything is installed.

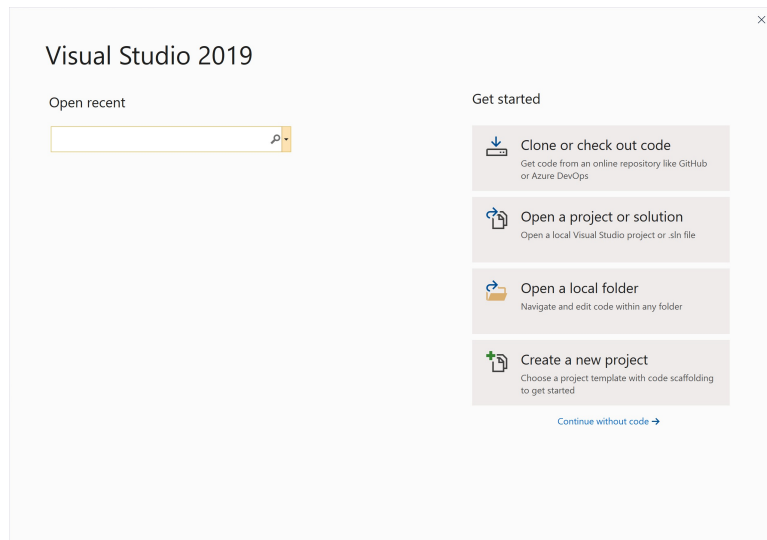
At this point, you should have a working version of the IDE installed. Let's check to make sure you do.

We put a shortcut on our desktop to make launching the IDE easier; for us, the IDE is installed at C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\Common7\IDE\devenv.exe.

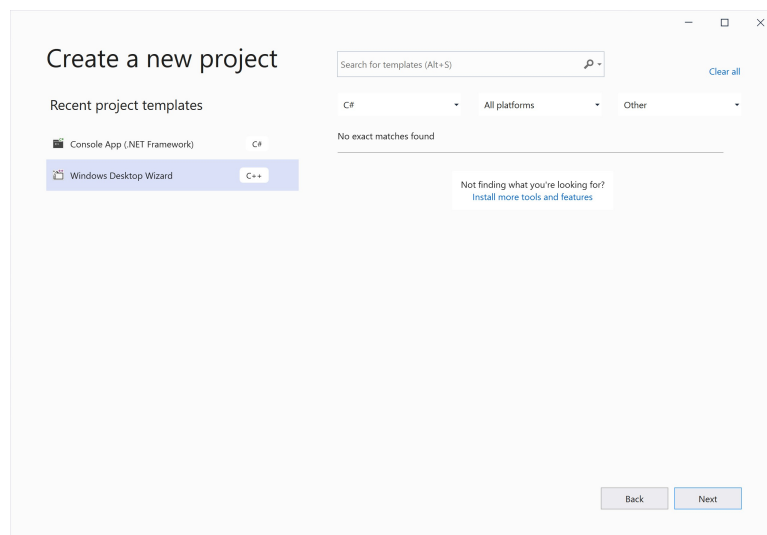
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Of course, you can also use Windows to search on Visual Studio and click the Visual Studio 2019 result instead if you prefer.

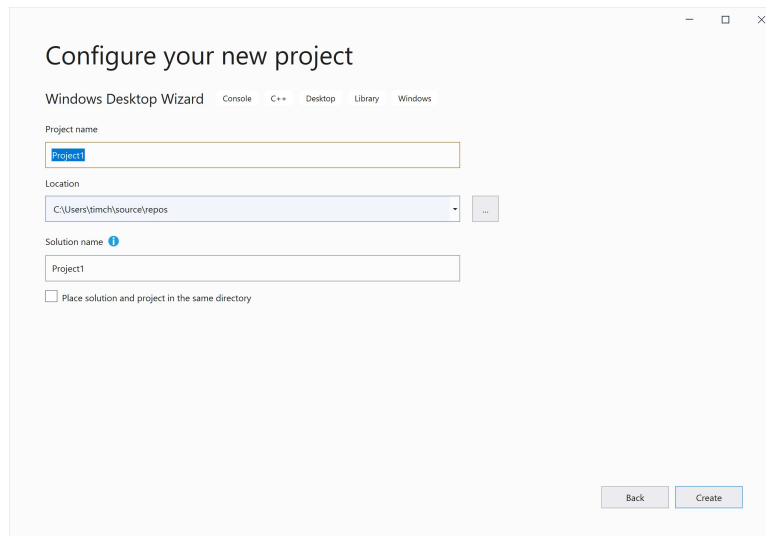
During the first startup of Visual Studio, be sure to change Development Settings to Visual C++ before clicking the Start Visual Studio button on the bottom right. When Visual Studio starts up, you should end up with something like the window below. Yours may not match exactly, but it should be similar.



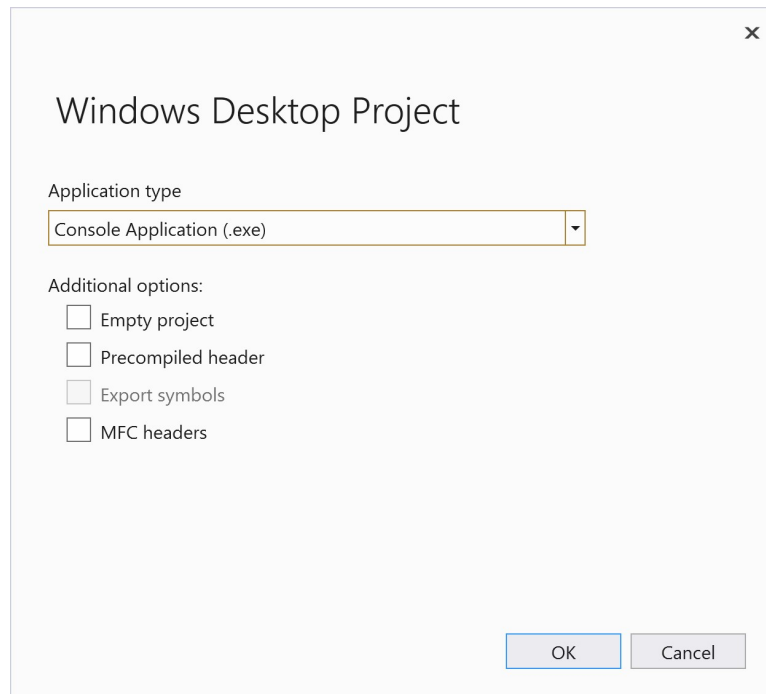
We're going to compile and run a simple program to make sure the IDE installed correctly. Click the Create a new project box on the lower right, which will bring you to a screen like the one below. Select the Windows Desktop Wizard C++ button on the upper left, then click the Next button on the lower right.



You'll end up at a screen like the one shown below. Change the Project name from ConsoleApp1 to InstallTest in the text box at the top left of the screen, change the Location to put the project wherever you want it to be saved, then click the Create button at the bottom right.

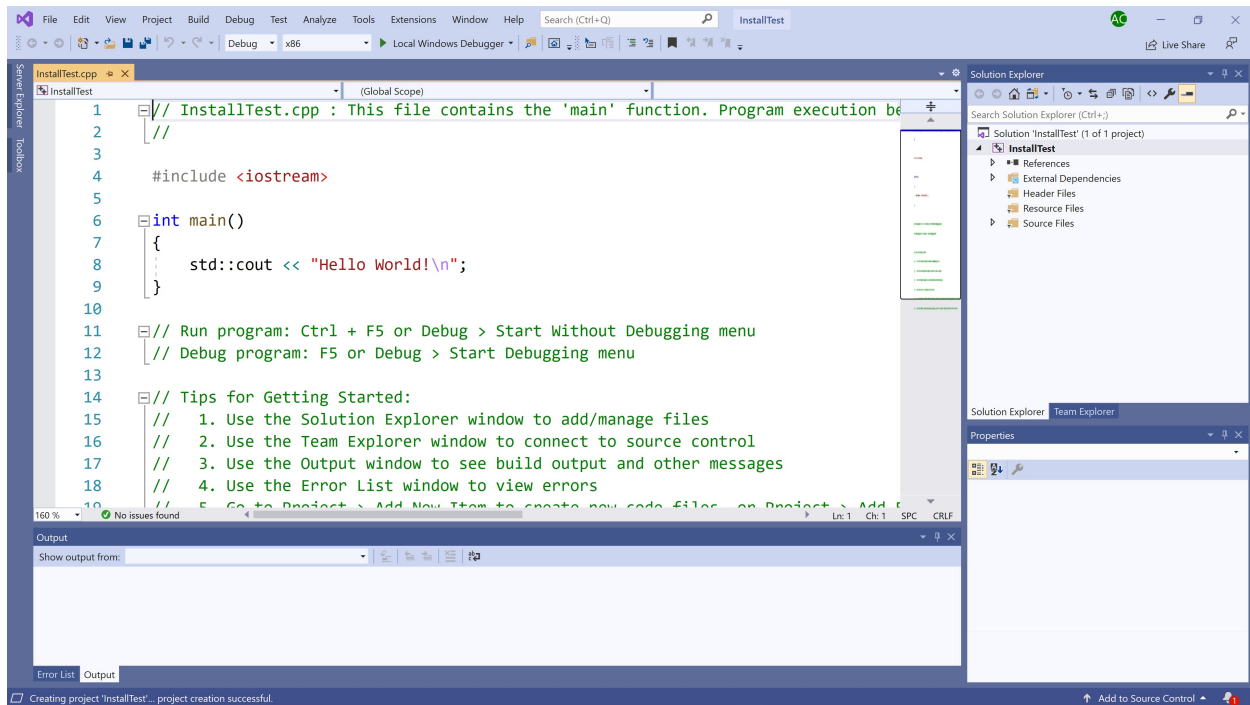


Next, you'll get a dialog as shown below; just click the OK button.



After a moment or two, you'll come back to the main screen of the IDE, which should look like this (we zoomed in on the Code window in the upper left using Ctrl + middle mouse wheel):

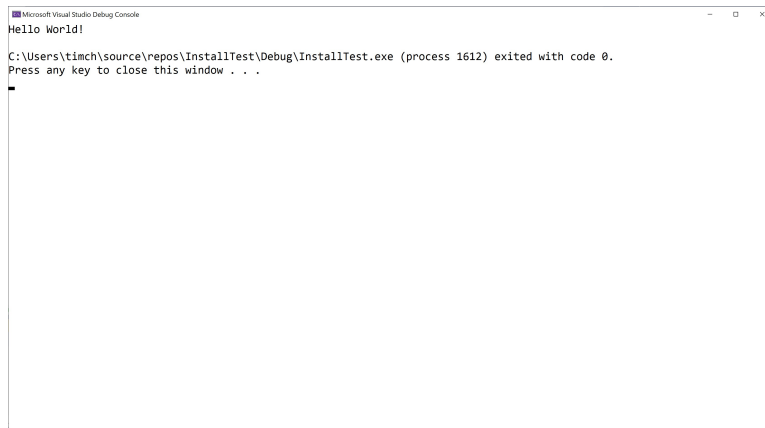
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If you have the Toolbox pane displayed on the left-hand side, you can simply click the Auto Hide "pin" just to left of the X in the upper right corner of the Toolbox pane to make it hide itself on the left.

When we told the IDE we wanted a new Console Application project (which I'll abbreviate to console app), it automatically gave us a template for that kind of application. Compile the application by pressing F6 or by selecting Build > Build Solution from the top menu bar¹. Once you see a "Build succeeded" message in the bar at the bottom of the IDE, you can actually run the application. Do that by pressing Ctrl+F5 (or selecting Debug > Start Without Debugging from the top menu bar); when you do, you should get the Command Prompt window below. Just press any key to return to the IDE.

¹ We've actually seen the keyboard shortcut for building the solution as F7 and Ctrl+Shift+B also. If you select Build > Build Solution from the top menu bar, the keyboard shortcut for that action is shown to the right. You can also customize the keyboard shortcuts in Visual Studio if you'd like; see <https://msdn.microsoft.com/en-us/library/5zwses53.aspx>



Your Command Prompt window probably has the default black background with light grey text on it rather than a white background with black text. Although that works fine on a computer screen, it looks horrible in an electronic or printed book! If that's the only difference between your output and the output above, you're good to go. If you decide you want to change your defaults, open a Command Prompt window (be sure to Run as administrator), right click near the top of the window, select Defaults, click the Colors tab, and change away. We also changed the Font, so feel free to customize that window as much as you want.

Exit the IDE. You certainly don't need to save the project if you don't want to – you'll get plenty of practice creating and saving projects – but feel free to do so if you want to.

If everything worked as described above, Visual Studio Community 2019 is installed and you're ready to move on to setting up the documentation. If your IDE isn't working, you should try working through the install process again or get online with Microsoft to ask for help.

When we're developing code, it's really useful to have the help documentation available directly within the IDE. The default for help documentation is to simply access help (through the IDE) online. If your computer always has connectivity to the Internet, this should work fine. If you're sometimes disconnected while you're programming, though, you'll probably want to set up your environment to use local help instead.

Start up the IDE, then click Help > Add and Remove Help Content. The IDE starts up the Microsoft Help Viewer, which lets you add local content. At a minimum, you should add the following topics:

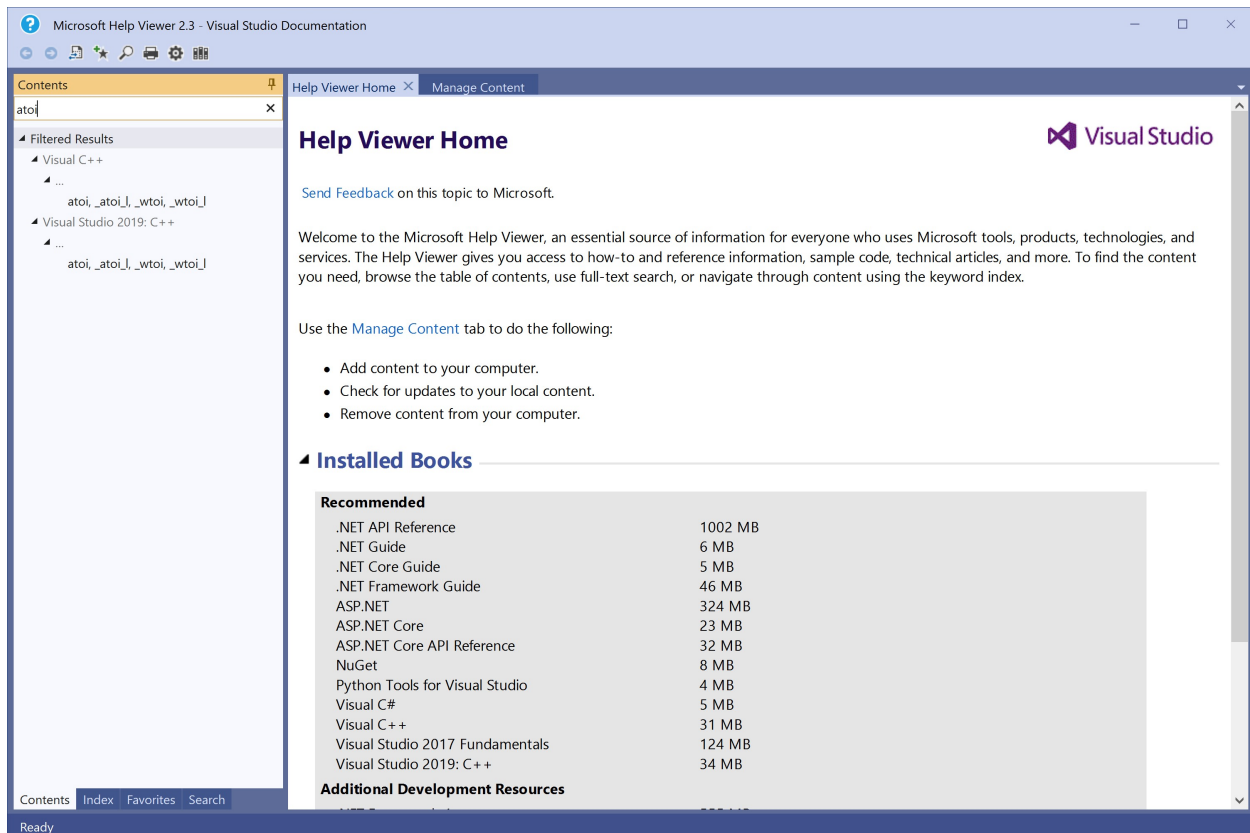
Recommended Documentation: Visual Studio 2019: C++ (already set to be added by default)
Available Documentation, Recommended: Visual C++ (already set to be added by default)

If the items above show that they're already installed but updates are available, click the download link near the bottom right. Click the Update button near the lower right of the Help Viewer and wait patiently while the update packages are installed.

Finally, you can read the help documentation through a browser (which will access the online documentation) or through the Microsoft Help Viewer (which will access the local help content).

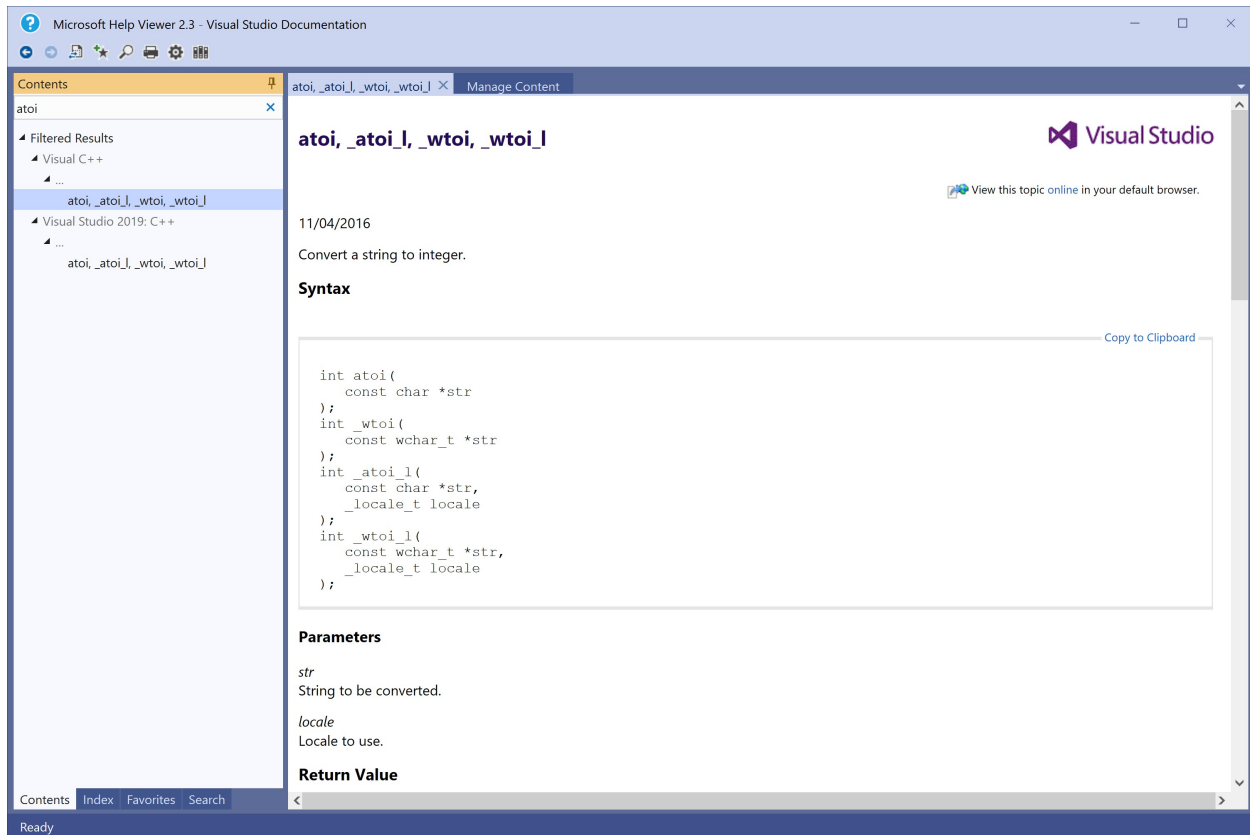
Both approaches provide access to the same help content, they just look slightly different – and, of course, if you're offline you can only access the local help. If you prefer using a browser, you don't need to make any changes because that's the default. If you prefer using the help viewer (like we do), select **Help > Set Help Preference > Launch in Help Viewer**.

Let's make sure the local documentation installed properly. Click **Help > View Help**. Enter `atoi` in the search box and press **Enter**. You should get a page of search results like the one shown below.



Your page will look different if you're using online help, but you should see a set of results somewhere on the page.

Click on the first link that says `atoi, _atoi_l, _wtoi, _stoi_l` and you should get a window of documentation like the one shown below.



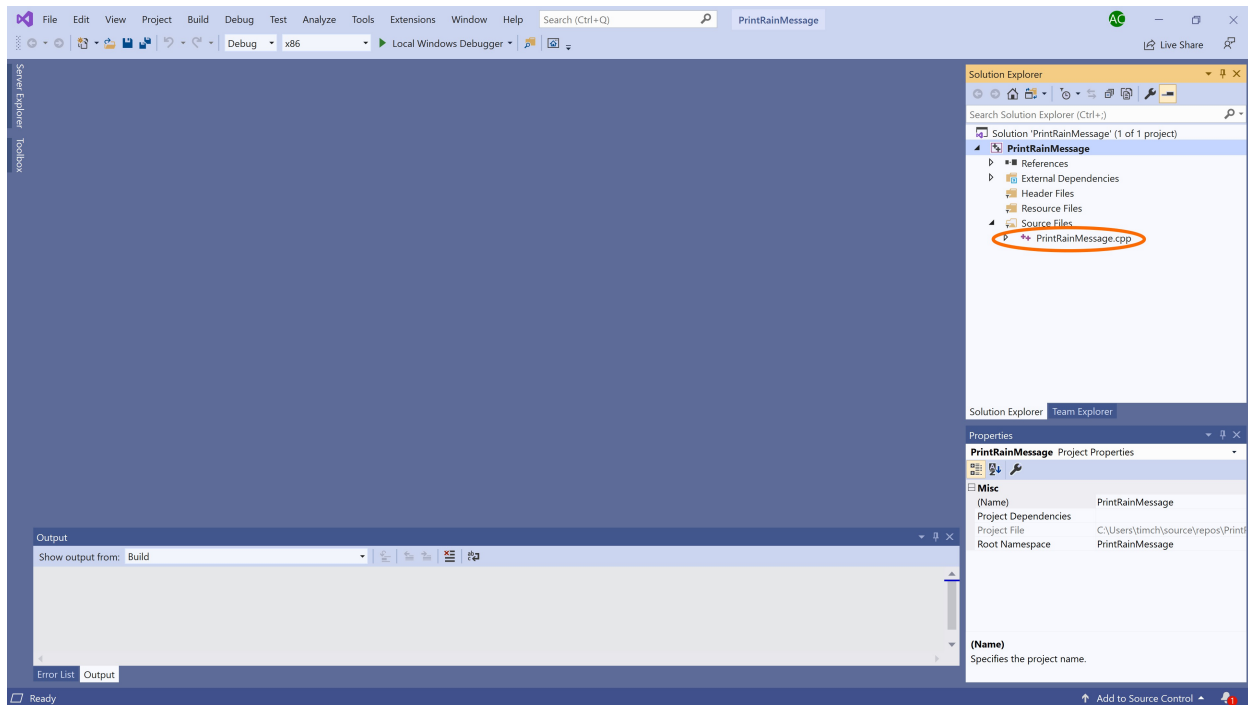
You should also know that you can always look at the C++ documentation online at <https://docs.microsoft.com/en-us/cpp/cpp/cpp-language-reference?view=vs-2019>; just enter your search term in the box in the upper right and press the enter key to look for a specific C++ topic. You might also find the [cplusplus.com reference](http://cplusplus.com/reference) useful. The bottom line is there are lots of C++ references you can use, so find one you're comfortable with. You'll be reading lots of C++ documentation as you write your C++ code!

Opening Downloaded Visual Studio Projects

When you download a Visual Studio project, you can open it in Visual Studio by double clicking the solution file. The solution file has a .sln file extension, but in case you're not showing file extensions in Windows Explorer it's the file marked "Microsoft Visual St...", "Visual Studio Solu...", or something similar as the Type. Let's see how that works.

Download the 1 2 4 Visual Studio Final Code zip file from the Resources for the Your First Console App in Visual Studio video in Week 1 of the Introduction to C++ Programming and Unreal course on Coursera from Coursera and extract the zip file somewhere on your computer.

Navigate into the folder called PrintRainMessage (it will be wherever you extracted the Chapter 2 code zip file) and double click the solution file. When the project opens, you might think something is wrong because the code pane doesn't show any code. Simply click PrintRainMessage.cpp in the Solution Explorer to see the code in that file.



That's all you need to get started with Visual Studio, so let's move on to Unreal.

A.2. Xcode (Mac only)

As noted above, if you're using a Mac you need to use Xcode as your C++ Integrated Development Environment (IDE). Here's how you can get Xcode installed:

1. Open Safari and search on xcode
2. Pick the first result (developer.apple.com)
3. Click the blue Download button near the top right
4. Sign in to Apple Developer if necessary. If you don't have an Apple Developer account, you should create one
5. Click the blue Download button on the right
6. Click through to get to the Xcode page, then click the cloud icon near the upper right to download, and wait patiently while Xcode is being downloaded
7. Close Safari
8. Click Finder, select Applications in the pane on the left, and scroll as necessary to find Xcode

The first time you run Xcode, you may be asked to install additional components; you should go ahead and do that.

Running Xcode Using Launchpad

To run the software using Launchpad, click the Launchpad icon in the dock as shown below.



Find the application you want to run (you may have to swipe to navigate through the applications you have installed) and click the icon for the application. The Xcode icon in Launchpad is as shown below.

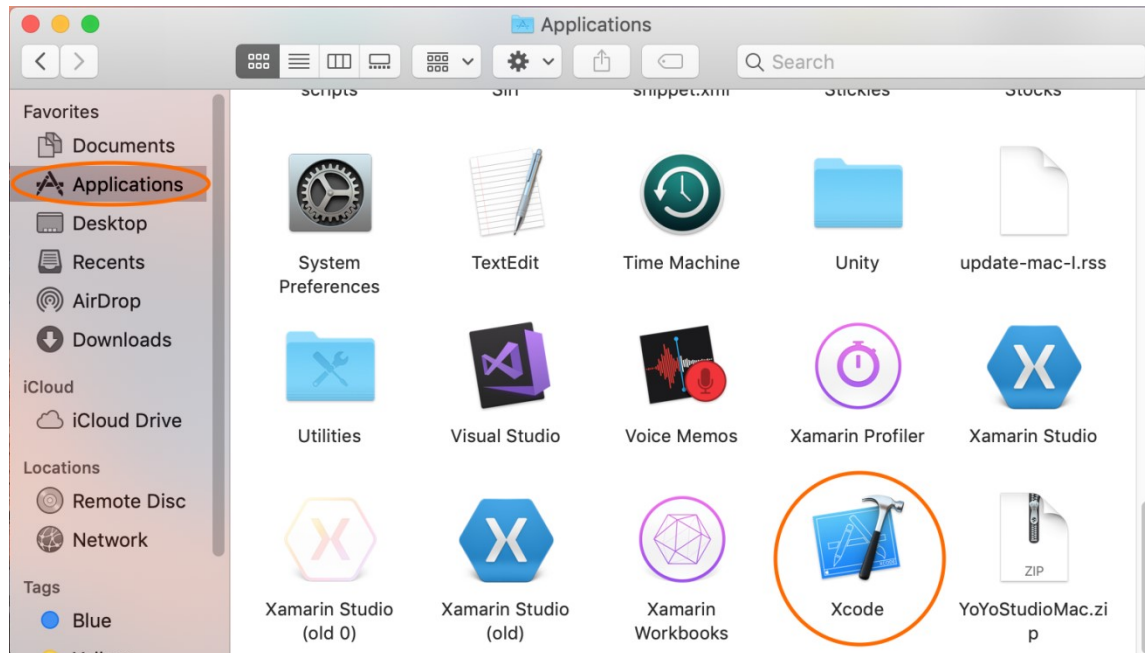


Running Xcode using Finder

To run the software using Finder, click the Finder icon in the dock as shown below.



Select Applications in the left pane of the Finder window and scroll as necessary to find Xcode. Click the icon for Xcode (see below) to run it.



Running from the Dock

For convenience, I just added Xcode to the dock as shown below. I can run Xcode by clicking on the appropriate icon in the dock.



To add Xcode to the dock, start up the Xcode using Launchpad or Finder, go to the Xcode icon in the dock, right click Options, and select Keep in Dock.

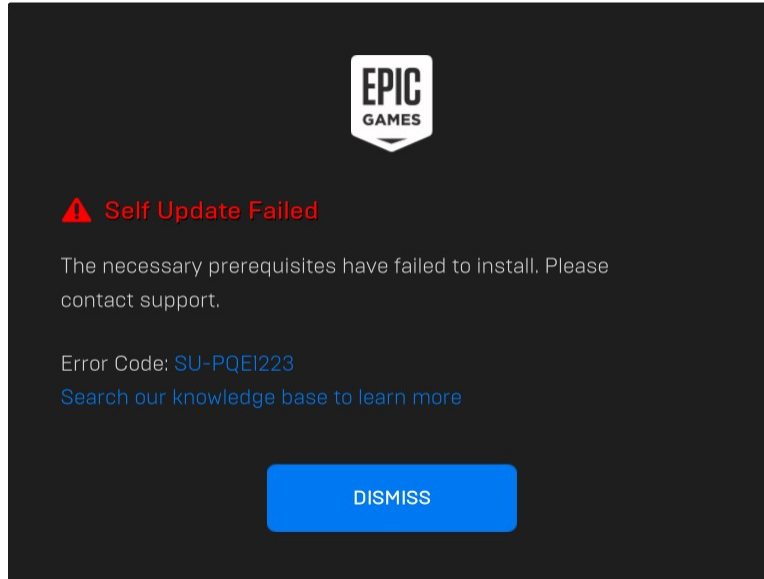
A.3. Unreal Engine

Be sure to install Visual Studio Community 2019 (Windows) or Xcode (Mac) before installing the Unreal Engine.

Here's how you can get the Unreal Engine installed:

1. Download the Unreal Engine from <http://unrealengine.com/>. At the time of this writing, you'll need to sign in with (or create) your Epic account, click Get Started Now, click Select under Publishing License, agree to the terms and conditions of the Publishing License as necessary, and wait for the download to complete. The sequence of steps may be different by the time you're doing this, but the bottom line is you're downloading the Unreal installer.
2. Run the install file you downloaded. Just accepting the default values as you go along should work fine, but you can of course change them if you want

When we installed the Epic Games Launcher, we got the error shown below.

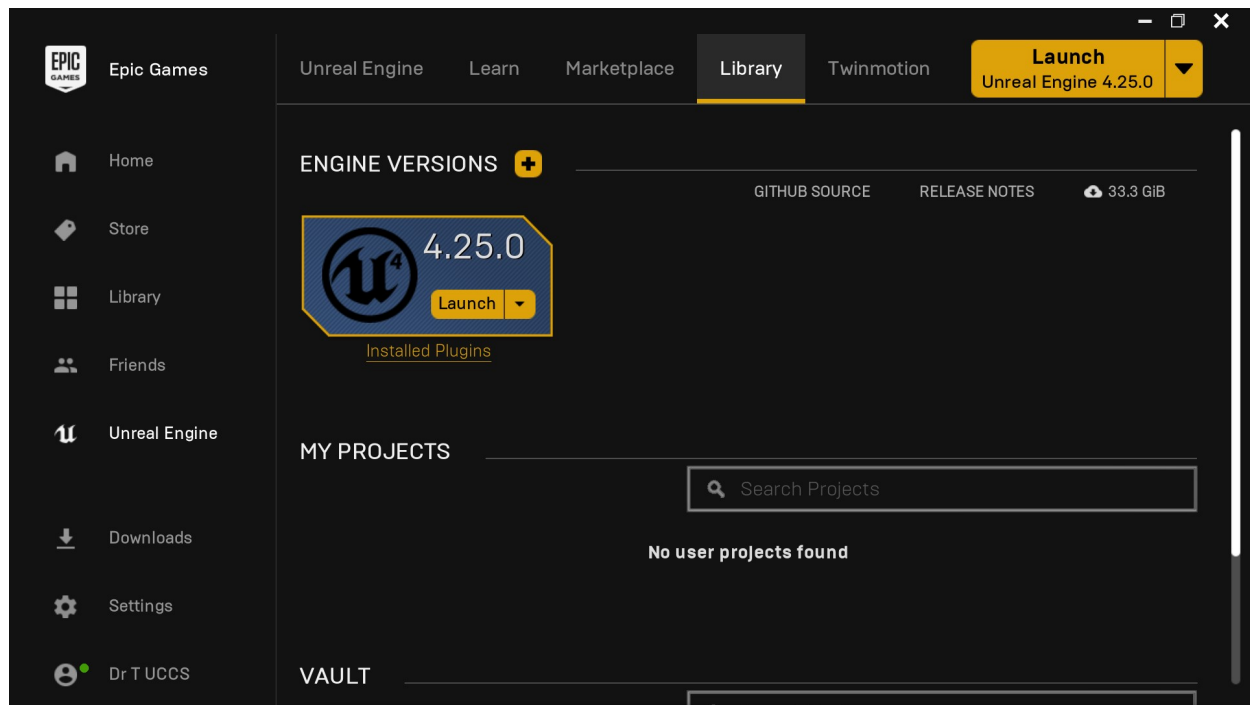


If you don't get that error, that's great, but if you do the following steps should solve this problem for you. Use your favorite search engine to search for "Microsoft Visual C++ 2010 Redistributable Package (x64)" (you should use (x32) instead if you have a 32-bit computer). Click on one of the top results hosted at microsoft.com and download and run the installer.

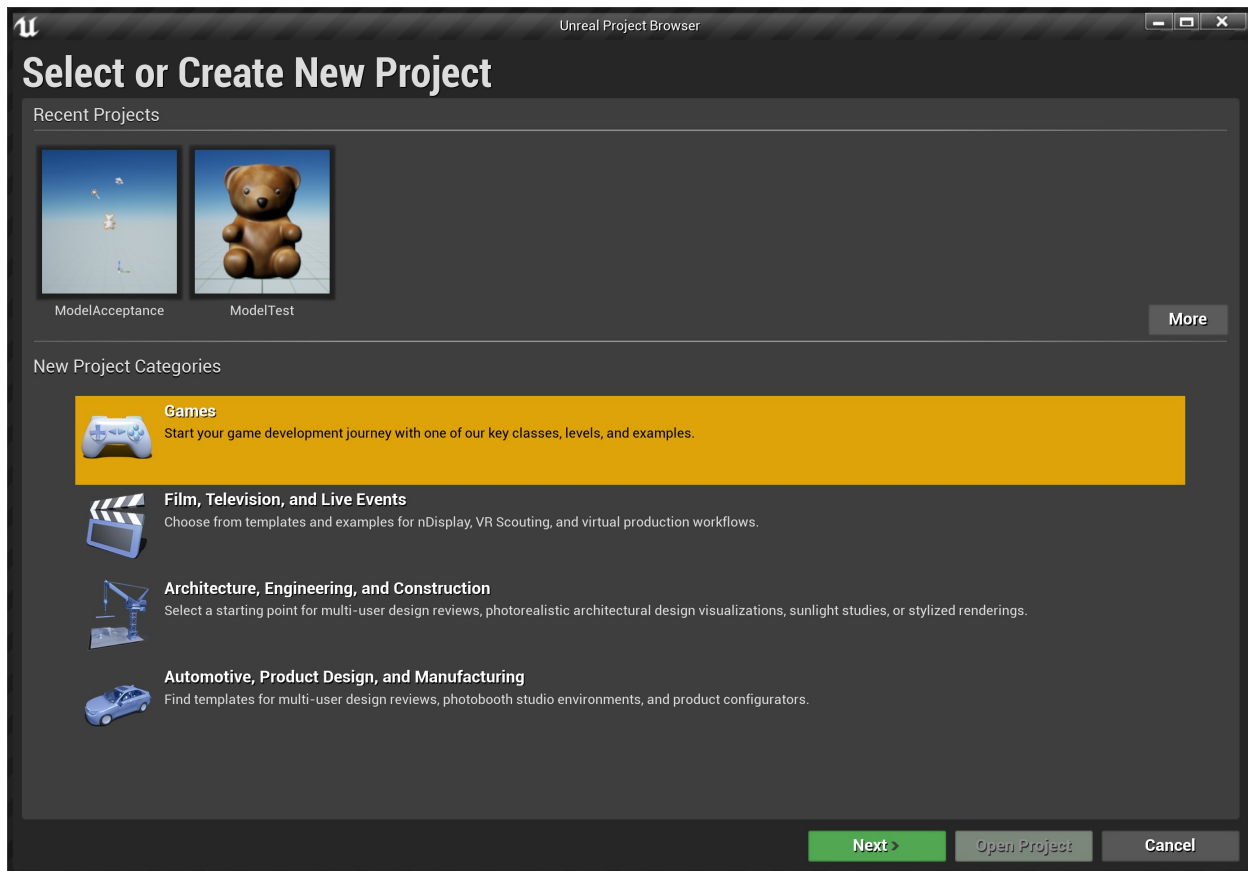
Now double click the Epic Games Launcher shortcut that was installed on your desktop and sign into your Epic account. Select Unreal Engine in the navigation bar on the left and click Install Engine near the upper right. Respond as necessary to dialogs and then wait while the engine is installed.

Note: When we recently installed the Epic Games Launcher in its default location, we couldn't install the Unreal Engine. This is a known problem that happens to some people. If this happens to you, try installing the Epic Games Launcher somewhere else; we just installed it at C:\Epic Games. We were then able to start up the Epic Games Launcher and install Unreal Engine.

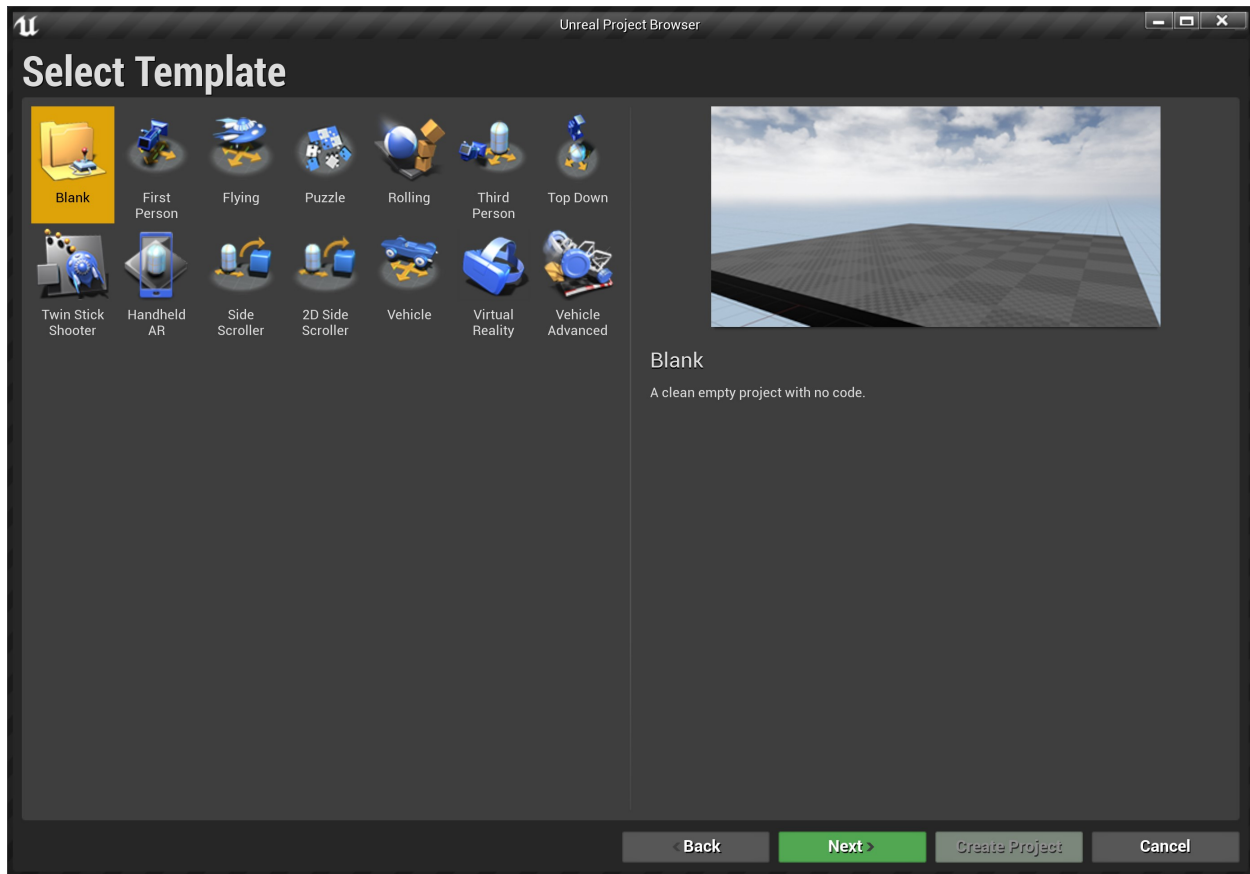
Once the Unreal Engine is installed, you should have something that looks like the figure below (though your Engine Version will be higher than 4.25.0).



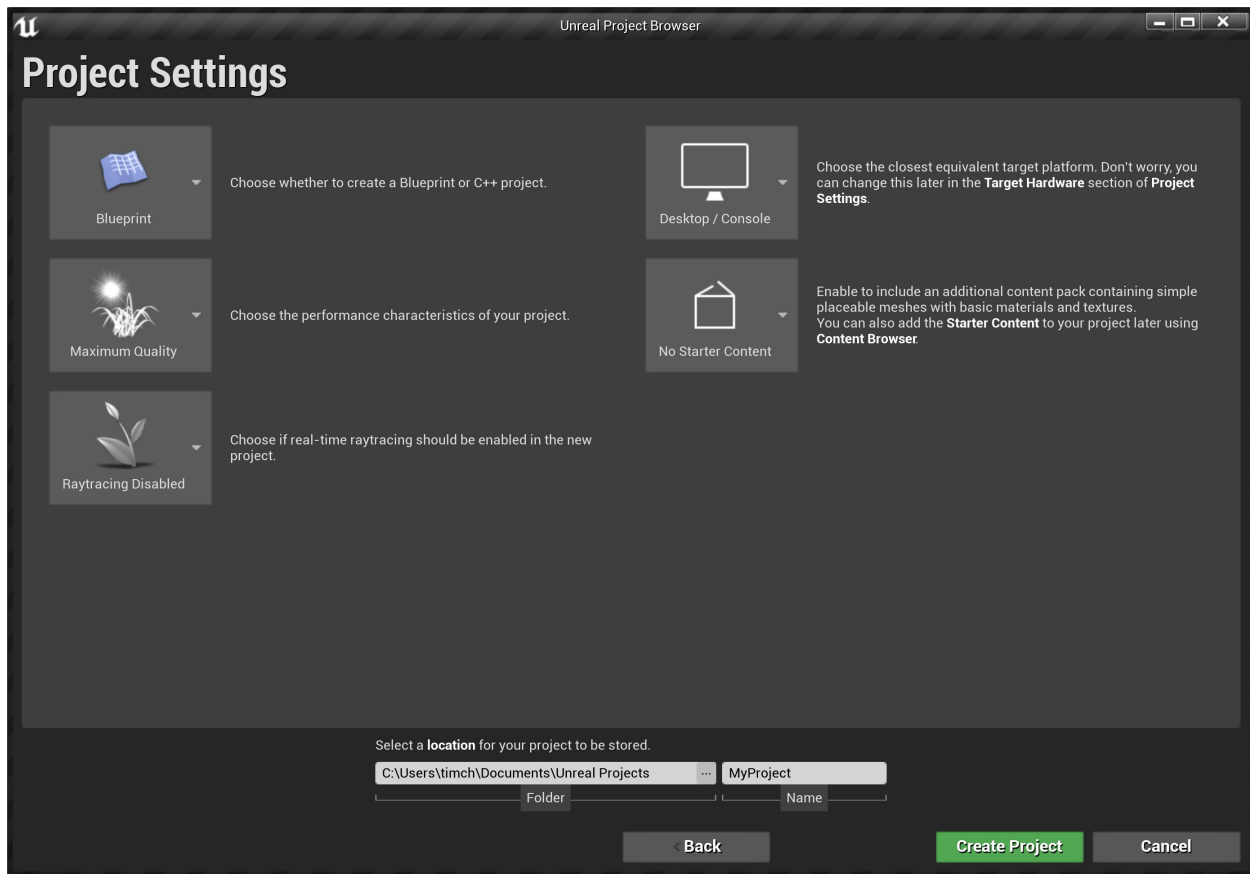
You can launch the Unreal Editor from here (by clicking either of the Launch buttons) or you can launch the Unreal Editor by double clicking the Unreal Engine shortcut that was installed on your desktop when you installed the engine. You should have something that looks like the figure below (without the Recent Projects, of course!).



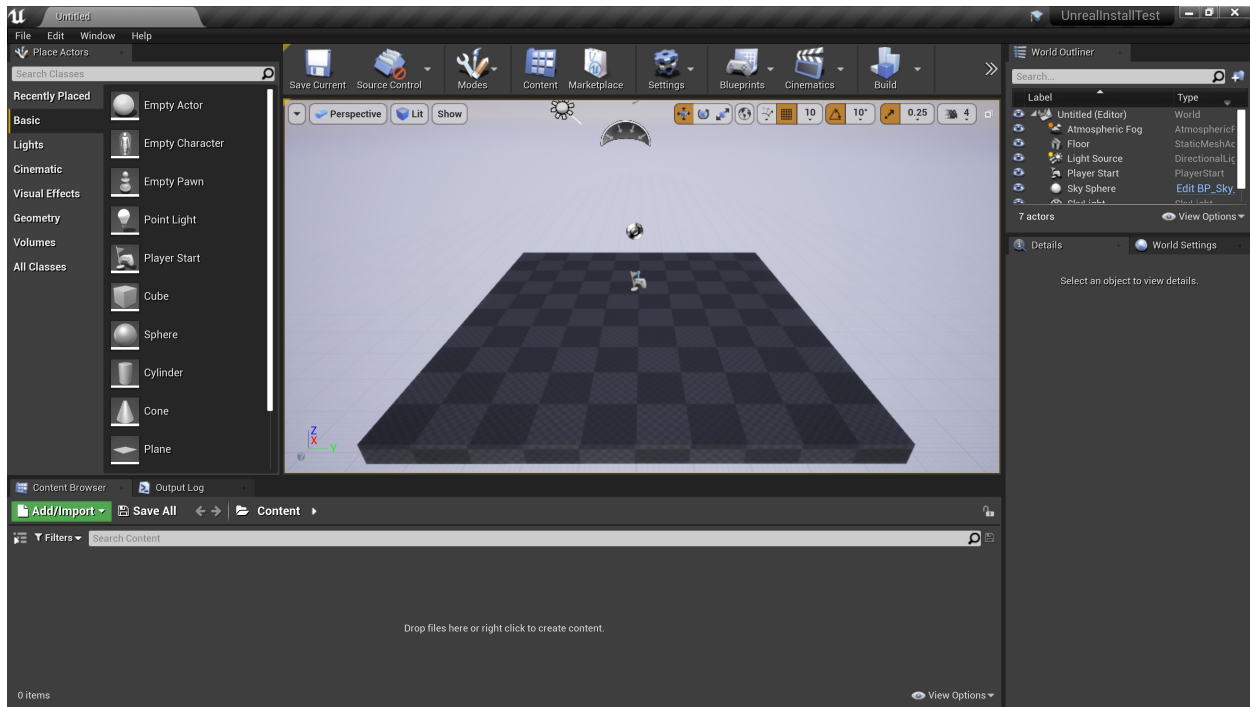
Select Games in the New Project Categories section, then click the green Next button near the lower right; you should have something that looks like the figure below.



Select the Blank project template near the upper right, then click the green Next button near the lower right. You should have something that looks like the figure below.



Click Blueprint in the upper left and change it to C++. Make sure the selection on the bottom of the column on the right says No Starter Content. In the middle near the bottom, change the Name to UnrealInstallTest, change the Folder to wherever you want to save the project, then click the green Create Project button at the lower right. After a moment of two, you'll come to the Unreal editor, which should look like this:



This is the default layout for the Unreal editor; although you can change the layout to suit your personal preferences, we'll just use the default layout.

Before we finish, let's make sure the Unreal Engine is set to use Visual Studio Community 2019 (or Xcode) as our script editor. Select **Edit > Editor Preferences ...** from the top menu bar, then click **Source Code** in the General section in the pane on the left. If the dropdown to the right of **Source Code Editor** in the Accessor section in the pane on the right DOESN'T say Visual Studio (or Xcode), click the dropdown and select Visual Studio (which will use the latest installed version of Visual Studio) or Visual Studio 2019 or Xcode (if you're on a Mac). Note: This is why we installed Visual Studio (or Xcode) before installing Unreal.

We'll actually add and test our first C++ script in Unreal in the appropriate Coursera course, so you're done setting up your development environment.

Opening Downloaded Unreal Projects

Download the Chapter6Code zip file from the Chapter 6: Unreal Engine Basics Reading in Week 4 of the Introduction to C++ Programming and Unreal course on Coursera and extract the zip file somewhere on your computer. Navigate into the WorstGameEver folder and double click the WorstGameEver.uproject file (the type is Unreal Engine Project File if you don't show file extensions) to open the project in the Unreal editor.

When you open the solution (or any Unreal projects you created), Unreal does NOT automatically open a level (map) for you. To open the level, type **Ctrl+O** (or go to the top menu bar and select **File > Open Level ...**). Double-click on Map0 to open the level.

Installing New Unreal Versions

Unreal releases new versions pretty regularly; you don't have to install each new version, but you'll probably want to install the latest version of Unreal periodically. Here's how you do that.

Open the Epic Game Launcher, select Unreal Engine on the left and select the Library tab on the top. Click the + button next to Engine Versions; if there's a newer version available, it will show up in the pane below. Click the Install button to install the new version.